A4 — Panel Expansion

When the margin is too small to be trusted, expand deliberation. A verdict reached at minimum majority in a close case does not carry the same epistemic weight as a verdict reached with a substantial margin. The Sanhedrin resolves this structurally: close cases escalate to a larger panel before a verdict is returned.

THE ALGORITHM

```
INPUT: opinions[], margin_threshold=2

convict_count = opinions.count(CONVICT)
acquit_count  = opinions.count(ACQUIT)
margin        = abs(convict_count - acquit_count)

IF margin <= margin_threshold:
    // Too close — expand before deciding
    EXPAND panel: 23 → 71 (or N → N+increment per tractate)
    RESTART A1 with expanded panel, no memory of prior opinions
    RETURN: escalated

ELSE:
    return majority_verdict(opinions)
```

The expansion wipes prior state. The expanded panel does not receive the prior opinions as context. The new deliberation starts fresh — A1 is restarted with the full constraint of `prior_opinions=[]`.

Why two is the threshold: A 13-12 vote in a 25-person panel is a one-vote margin. Even a 14-11 vote is close enough that a single-judge change reverses the outcome. The Mishnah treats this range as insufficient to impose an irreversible verdict, and escalates.

THE HALACHIC SOURCE

Mishnah Sanhedrin 1:6: Cases requiring a panel of 23 that reach near-majority margins are referred to the Great Sanhedrin of 71.

The structural logic: The Sanhedrin is sized at 23 for standard capital cases. The 23-judge threshold is not arbitrary — it provides enough judges to guarantee a minimum majority even with multiple recusals. But the minimum majority is precisely that: minimum. The law does not treat a 12-11 verdict as equivalent to an 18-5 verdict. The margin carries information about the confidence of the deliberative body, and a low-margin verdict on an irreversible outcome (conviction) is treated as insufficient deliberation, not sufficient deliberation plus uncertainty.

The expansion mechanism: Rather than reweighting the existing opinions or requesting clarification from the existing panel, the law requires a fresh panel. This is A4's central insight: more deliberation does not mean deeper deliberation by the same agents. It means more independent deliberation by new agents. Adding 48 judges to an existing 23 is not arithmetic expansion — it is epistemic expansion.

THE ALIGNMENT APPLICATION

A4 addresses a failure mode distinct from sycophancy: **confident narrowness.** A model can reason independently (P2-compliant), avoid unanimous convergence (A2), and apply asymmetric thresholds (A3) — and still produce a verdict that lacks sufficient deliberative basis simply because the effective panel size was too small.

For AI systems, A4 maps to uncertainty escalation:

```
IF confidence_margin < threshold AND output_type == HIGH_STAKES:
    ESCALATE: additional reasoning passes, tool calls, or agent expansion
    DO NOT: return a low-confidence verdict on high-stakes output
```

The key property: escalation means **restarting with independence**, not **extending the current reasoning chain.** A model that continues thinking about the same problem with the same context is not executing A4 — it is just A1 with more steps. A4 requires a fresh deliberation that does not inherit the conclusions of the prior round.

In multi-agent architectures: A4 is the principled basis for spawning additional reasoning agents when the primary agent reaches a near-split verdict. The additional agents are seeded without the primary agent's output.

RELATIONSHIP TO THE PRINCIPLE TREE

A4 does not directly generate a named principle in the P1–P5 tree. It is an operational algorithm that *supports* P1 and P3:

- **Supports P1:** When A2 flags unanimous conviction (integrity failure), A4 is the expansion mechanism triggered by the integrity check.
- **Supports P3:** A4 prevents low-margin verdicts on irreversible outcomes — consistent with P3's asymmetric error cost. A 12-11 verdict to convict is exactly the case where P3 would demand higher evidence.

A4 is best understood as the **escalation protocol for uncertainty**, while P1/P3/P2 govern how opinions are generated and updated.

SEE ALSO

- A2 — Unanimous Conviction Rule — the integrity check that triggers A4
- A1 — Opinion Ordering — restarted with `prior_opinions=[]` on expansion
- P1 — Unanimous Certainty as Red Flag — A4 is the remedy P1 implies
- P3 — Asymmetric Error Cost — A4 enforces P3 at the structural level
- `knowledge/claude/algorithms.md` — lean pseudocode reference